

TF-IDF Methodology and Results

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Abstract

This contribution examines word-based authorship attribution for Reuters articles by William Kazer, Tim Farrand and Patricia Commins. It combines TF-IDF representations, principal component analysis, hierarchical clustering and a nearest-neighbour demonstration. The results illustrate lexical differences between the selected authors, while also exposing the difficulty of separating writing style from subject matter.

1 Corpus and experimental setup

The experiment uses the `Reuter_50_50` corpus [1]. Fifty articles per author are read from `C50train`. Within each author’s alphabetically sorted file list, the first article is reserved as a mystery text. The remaining 49 articles per author form a reference set of 147 documents. The vectoriser and PCA are fitted on this reference set. The three mystery texts are subsequently transformed with the fitted objects.

This is a small exploratory holdout experiment within the original training partition. The three mystery texts are not the corpus’s official test set. The complementary style-based contribution evaluates the official test articles separately.

2 TF-IDF vectorisation

The Reuters notebook and the shared `stylometry/features.py` module use a custom regular-expression tokenizer and NLTK’s English Snowball stemmer [2]. Stemming reduces some inflectional variants, such as *company* and *companies*, to a shared representation. It is an implementation choice, not a necessary property of journalistic texts. Nor does stemming guarantee that every semantically related noun and verb receives the same stem.

The `TfidfVectorizer` settings are `max_features=1000`, `min_df=2`, `max_df=0.9`, English stop words, and word n-grams of length one to three [3]. The selected vocabulary therefore contains stemmed unigrams and multi-token sequences, not 1,000 distinct single-word stems. The sparse matrix returned by the vectoriser is converted to a dense array in the notebook.

With the default smoothed inverse document frequency, the weight before normalisation is

$$w_{d,t} = \text{tf}(t, d) \left[\log \left(\frac{1 + N}{1 + \text{df}(t)} \right) + 1 \right].$$

Here, $N = 147$, $\text{tf}(t, d)$ is the feature count in document d , and $\text{df}(t)$ is its reference-document frequency. Each document vector is normalised to unit Euclidean length. The English stop-word

list and the custom stemmer are not fully aligned, which produces a scikit-learn warning. The verification retains the notebook’s settings so that the comparison does not silently change its preprocessing.

3 Principal component analysis

PCA projects the 1,000-dimensional reference vectors onto two axes for visualisation. The original report gives 6.20% for PC1 and 5.35% for PC2, or 11.55% together. A local check using an exact full-SVD solver gives 6.23% and 5.35%, or 11.58% together. Thus, most variance lies outside the displayed plane. A visually distinct group in this projection should not be treated as proof of reliable classification.

The author-average TF-IDF profiles provide a concrete interpretation of the lexical differences. These average weights are different from PCA loadings, which describe the directions of the projection.

Author	Feature	Reported mean	Rechecked mean
William Kazer	china	0.1912	0.1914
Tim Farrand	million	0.1512	0.1516
Patricia Commins	adm	0.1236	0.1231

The reported profiles associate Kazer with China and trade, Farrand with financial quantities, and Commins with particular companies and business reporting. These are primarily topical or lexical clues. They cannot be interpreted as independent evidence of a stable personal writing style.

The Federalist notebooks offer a historical comparison, but their corpus and preprocessing differ. A difference in two-component explained variance does not, by itself, establish that one corpus has a more homogeneous vocabulary than the other.

4 Hierarchical cluster analysis

The Reuters notebook computes pairwise Euclidean distances between the full TF-IDF reference vectors and applies Ward linkage to the condensed distance vector [4]. It does not cluster the two-dimensional PCA coordinates. Ward linkage successively merges groups with a small increase in within-cluster squared deviation.

The reported dendrogram was interpreted as showing a comparatively distinct Farrand group and some overlap between Kazer and Commins. Such structure can reflect shared topics as well as author differences. The dendrogram is therefore an exploratory description of similarity, not an independent measure of authorship accuracy.

5 Nearest-neighbour demonstration

For each mystery document, the notebook calculates its Euclidean distance to every reference vector and predicts the label of the closest article:

$$\hat{y}(x) = y_{i^*}, \quad i^* = \arg \min_{i \in \mathcal{R}} \|x - x_i\|_2.$$

This classification also uses the full TF-IDF space. PCA is used for display only.

True author	Prediction	Reported distance	Rechecked distance
William Kazer	Kazer	1.1217	1.1217
Tim Farrand	Farrand	0.8650	0.8615
Patricia Commins	Commins	0.3407	0.3407

All three mystery texts are classified correctly, giving an accuracy of 3/3 for this particular selection. Commins’s mystery text has the closest lexical neighbour. A low distance can also reflect topic or event similarity, rather than a general author-specific consistency.

The original values are retained alongside the local check from 4 October 2026. The check records its software versions and document identifiers. The small numerical differences have not been traced to a single cause. The three predicted authors agree in both runs.

Three correctly classified texts do not establish robustness across new articles. The mystery documents are selected deterministically rather than repeatedly resampled. The complementary contribution evaluates the official test partition and reports 137 correct TF-IDF predictions out of 150 articles.

6 Relation to the style-based contribution

This analysis supplies a word-based reference for the joint project. The complementary contribution preserves the selected authors and reference articles, then compares explicit style features with the same nearest-neighbour rule. It also adds an author-centroid baseline and an evaluation on the official test partition. This connects exploration and evaluation without treating visible PCA clusters as classification results.

7 Implementation and reproducibility

The repository provides `prepare_data.py` for corpus preparation and `run_analysis.py` for the complete comparison. The Reuters notebook imports the shared tokenizer and vectoriser, then calculates HCA, PCA and mystery predictions in separate cells. A fresh Python 3.13.13 installation reproduced the reference results. The supplementary Federalist notebooks now share a body-only parser, consistent disputed/joint categories and correct Ward input. Their revised plots replace the earlier uncleaned exploration. The shared report documents the full evaluation and project responsibilities.

References

- [1] Liu, Z. (2006). *Reuter_50_50*. UCI Machine Learning Repository. [doi:10.24432/C5DS42](https://doi.org/10.24432/C5DS42).
- [2] NLTK. *English Snowball stemmer*. [Online documentation](#).
- [3] scikit-learn. *TfidfVectorizer*. [Online documentation](#).
- [4] SciPy. *Hierarchical clustering: linkage*. [Online documentation](#).